WORKED SOLUTION

STEP I 1994, Question 13

Probability and Statistics • Indicator Variables

> **STEP I 1994, Question 13**
>
> Read the question before you go on. It is in the STEP Support Programme's question database:
>
> step.maths.org/questions/94-s1-q13

The key idea

Writing a count as a sum of indicator variables (each 1 or 0) turns a hard expectation into an easy one, because expectation is linear *whether or not* the pieces are independent. Here the draws are certainly not independent, but symmetry tells us each X_i has the same expectation. Then we compute $\mathrm{E}(X)$ a second way, straight from the distribution of X, and setting the two answers equal gives a binomial-coefficient identity almost for free.

Solution

Throughout, $1 \le n \le M$, and we use the convention that $\binom{a}{b} = 0$ when $b > a$ or $b < 0$.

Part (i)

Each X_i is 1 if the ith token removed is red and 0 if it is not. So in the sum $X_1 + X_2 + \cdots + X_n$ every red token removed contributes exactly 1 and every non-red token contributes 0. The sum therefore counts the red tokens among the n removed, which is X.

Part (ii)

Since X_i takes only the values 0 and 1,

$$\mathrm{E}(X_i) = 0 \cdot \mathrm{P}(X_i = 0) + 1 \cdot \mathrm{P}(X_i = 1) = \mathrm{P}(i\text{th token removed is red}).$$

> **Ask yourself** Doesn't the chance that the ith token is red depend on what was drawn before it?

Conditionally, yes; unconditionally, no. Imagine numbering the tokens and drawing them all out one by one: every ordering of the M tokens is equally likely. So the token in position i is equally likely to be any of the M tokens, just as the token in position 1

is. Stopping after n draws doesn't change what happens at position $i \le n$. Since m of the M tokens are red,

$$\mathrm{E}(X_i) = \mathrm{P}(X_i = 1) = \frac{m}{M} \qquad \text{for each } i = 1, \ldots, n.$$

(For $i = 2$ you can confirm this the long way: $\frac{m}{M} \cdot \frac{m-1}{M-1} + \frac{M-m}{M} \cdot \frac{m}{M-1} = \frac{m(M-1)}{M(M-1)} = \frac{m}{M}$.)

Part (iii)

By part (i) and linearity of expectation (which needs no independence),

$$\mathrm{E}(X) = \mathrm{E}(X_1) + \cdots + \mathrm{E}(X_n) = n \cdot \frac{m}{M} = \frac{mn}{M},$$

as required.

Part (iv)

The set of n tokens removed is equally likely to be any of the $\binom{M}{n}$ subsets of size n. To have exactly k red tokens we choose k of the m red tokens and $n-k$ of the $M-m$ others, so

$$\mathrm{P}(X = k) = \frac{\binom{m}{k}\binom{M-m}{n-k}}{\binom{M}{n}}, \qquad k = 0, 1, \ldots, n.$$

The convention on binomial coefficients makes this correctly 0 when $k > m$ (not enough red tokens) or $n - k > M - m$ (not enough others). This is the *hypergeometric* distribution.

Part (v)

Compute $\mathrm{E}(X)$ directly from part (iv); the $k = 0$ term is zero, so

$$\mathrm{E}(X) = \sum_{k=1}^{n} k\,\mathrm{P}(X = k) = \frac{1}{\binom{M}{n}} \sum_{k=1}^{n} k \binom{m}{k}\binom{M-m}{n-k}.$$

Equating this with part (iii),

$$\sum_{k=1}^{n} k \binom{m}{k}\binom{M-m}{n-k} = \frac{mn}{M}\binom{M}{n}.$$

Finally,

$$\frac{n}{M}\binom{M}{n} = \frac{n}{M} \cdot \frac{M!}{n!\,(M-n)!} = \frac{(M-1)!}{(n-1)!\,(M-n)!} = \binom{M-1}{n-1},$$

so the right-hand side is $m\binom{M-1}{n-1}$, which is the required identity.

Checks

- Take $M = 5$, $m = 2$, $n = 3$. Part (iv) gives $\mathrm{P}(X = 0, 1, 2) = \frac{1}{10}, \frac{6}{10}, \frac{3}{10}$, which sum to 1, and $\mathrm{E}(X) = \frac{6+6}{10} = \frac{6}{5} = \frac{mn}{M}$. The identity reads $1 \cdot 2 \cdot 3 + 2 \cdot 1 \cdot 3 = 12 = 2\binom{4}{2}$.

- If $n = M$ (take everything), $X = m$ for certain, and indeed $\mathrm{E}(X) = mM/M = m$.
- We checked $\mathrm{E}(X_i)$, $\mathrm{E}(X)$ and $\mathrm{P}(X = k)$ by listing every ordering of the tokens for all bags with $M \leq 6$, and the identity in (v) for every $M < 30$.

Common slips

- Claiming $\mathrm{E}(X) = \sum \mathrm{E}(X_i)$ needs independence. It doesn't: linearity of expectation always holds.
- Treating the draws as independent and writing a binomial $\mathrm{B}(n, \frac{m}{M})$ in (iv). This gets the mean right but the probabilities wrong, because the draws are without replacement.

Beyond STEP

The indicator method is one of the most useful tools in probability: it gives expected numbers of fixed points of a random permutation, of runs in coin tosses, of isolated vertices in a random graph, and it extends to variances via $\mathrm{Var}(\sum X_i) = \sum \mathrm{Var}(X_i) + \sum_{i \neq j} \mathrm{Cov}(X_i, X_j)$. Proving combinatorial identities by computing one quantity in two ways is just as general. Blitzstein and Hwang, *Introduction to Probability*, uses indicators throughout and is an excellent place to see both ideas in action.

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